TensorFlow and Keras for AI in Fluids

Week 1, notebook 02 - a complete student companion

FlowMLLab course website

Start the lab: Run notebook 02 in Colab or read its current GitHub version.

This guide accompanies the existing Week 1 TensorFlow notebook 02. The course also contains a separate pressure-velocity lab called Week 1.2. Here, notebook 02 means the introductory TensorFlow lab.

The first part explains every worked section in notebook order. The second part turns the small sine-function network into a controlled laboratory. You will change data counts, network architecture and training settings, then explain the resulting predictions and loss curves.

Work with the notebook and this guide side by side. Run all cells once in a fresh kernel before changing the configuration. Keep a baseline and a short experiment record. State what you expected, what you changed, what you measured and what the evidence supports.

The field examples are manufactured analytic functions. The sine network learns a known function on a specified interval. The convolutional example checks tensor shapes using random inputs. These examples establish the programming and evaluation methods needed before learning from computed or measured flow fields.

Learning goals

Construct a correctly shaped tensor, check an automatic derivative, train a small neural network and compare training with validation fairly. Explain a loss change using a controlled experiment rather than attributing it to a cause from the curve alone.

What to submit

Submit the executed notebook, a compact experiment table and figures showing predictions and loss histories. Report the analytic reference, data counts and selected model. Evaluate the held-out test set after model selection is complete.

Guide map

Use these links to move through the guide.

- Sections 1-4: setup, tensors, broadcasting and channels
- Sections 5-8: losses, automatic differentiation and networks
- Sections 9-12: data splits, training and loss diagnostics
- Sections 13-16: data-count, architecture and activation experiments
- Sections 17-20: physical checks, convolution and submission

1. Read the notebook as an experiment

Notebook Section 0

The setup imports NumPy, Matplotlib and TensorFlow. NumPy supplies arrays and sample coordinates. Matplotlib displays fields and profiles. TensorFlow evaluates differentiable computations and stores trainable parameters. Keras supplies model, layer and optimizer interfaces.

The cell prints the TensorFlow version and available devices. A central processing unit is sufficient for this lab. A graphics processing unit can change timing and numerical details. Save the library version and device information with your results.

A notebook depends on its kernel state. Running a plotting cell before the cell that creates its inputs can produce a missing name. Editing the data without retraining can combine old weights with new labels. Restarting the kernel and running from the top checks the computation you intend to submit.

A saved Google Drive copy is a separate document. Open the current GitHub or Colab link when checking a course update. The notebook revision should agree with the material you are reading.

Check every cell

Write down input shape, output shape and represented quantity. An array with shape (121, 1) can represent 121 scalar coordinates. It could also contain 121 scalar temperatures. Shape alone supplies neither physical meaning nor units.

Identify values generated from an analytic formula, trainable values and evaluation data. A lower loss is meaningful only after you know which data and model state produced it.

First independent checks

Run the field cell and inspect its shape. Calculate the first mean squared error by hand. Run the scalar derivative and verify that the derivative is six. These checks distinguish installation problems from mathematical or data-handling problems.

Item	Record
Environment	Python and TensorFlow versions
Device	CPU or accelerator
Configuration	Counts, seed and settings
Evidence	Executed cells and figures
Interpretation	Claim tied to a comparison

2. Tensors describe the arrangement of data

Notebook Sections 1 and 3

Reference: TensorFlow tensor guide.

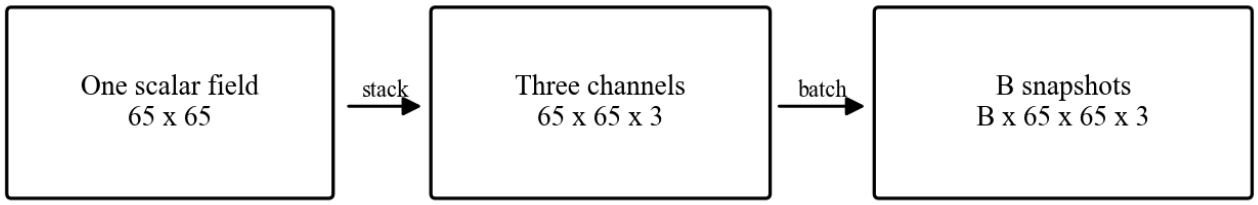
A TensorFlow tensor is a multidimensional array. Rank counts its axes. Shape gives the length of each axis. A scalar has shape (), a three-entry vector has shape (3,) and a square field on a 65-by-65 grid has shape (65, 65).

The notebook converts a NumPy array into a TensorFlow constant using float32, a representation with 32 bits per number. Float64 uses 64 bits. More precision changes storage and arithmetic cost. It cannot recover information absent from measurements or a numerical solution.

tf.linspace and tf.meshgrid construct the field coordinates. With indexing="xy", X and Y have shape (65, 65). The first index selects a y-row. Thus phi[10, :] follows a horizontal line, while phi[:, 10] follows a vertical line.

The manufactured field is phi = sin(pi x) sin(pi y). Its values are zero on the unit-square boundary and one at the centre. The 65-point grid includes that centre. Check these facts in the contour figure.

TensorFlow evaluates sine and multiplication over the arrays. For h = f + 0.1 g, where g = X squared + Y squared, all arrays have matching shapes. Printing minima and maxima checks the range but does not validate every coordinate.



Axis meanings travel with the data: batch, y, x, channel.

Field-data shapes

Try it

Change N to 33, then 129. Predict the field shape and number of entries before running. The function is unchanged while its sampling changes. Explain why doubling points in both directions roughly quadruples the number of entries.

The data of one float32 scalar field require 4 times Ny times Nx bytes. Coordinate arrays, plotting buffers and TensorFlow objects require additional memory.

3. Variables, broadcasting and a silent shape error

Notebook Sections 2 and 4

A tf.constant has a fixed value after construction. A tf.Variable stores a value that can be updated. The notebook starts w at 0.5 and uses w.assign(2.0). The same variable now stores 2.0.

A model's weights and biases are variables. Input coordinates and labels remain fixed for a training experiment. A CFD state can also be updated, but mutability alone does not make it a learned parameter.

Broadcasting lets a smaller tensor participate in an operation with a larger tensor. Shapes are compared from their trailing axes. Corresponding lengths must match or one must be one.

Multiplying the notebook's (4, 5) field by a scalar produces a (4, 5) result. Every entry receives the same scale.

The next cell demonstrates a silent error. Targets with shape (4, 1) and predictions with shape (4,) produce a (4, 4) residual on subtraction. Every prediction is compared with every target. The mean squared error can look plausible although the intended pairing has been lost.

predictions_1d[:, None] changes the prediction shape to (4, 1). The residual now has one entry per matched sample. In the worked example, the correctly paired residual is zero.

Diagnose before training

Print input, label and prediction shapes before calculating a loss. For scalar regression, use inputs shaped (N, features), with labels and predictions both shaped (N, 1). Check their shapes explicitly.

Try it

Repeat with five samples. Explain why the incorrect residual contains 25 entries and the correct residual contains five. Change a single prediction and identify which correctly paired residual should change.

A calculation that executes has passed a programming check. Coordinate alignment, units and physical meaning require further checks.

4. Channels retain their physical meaning

Notebook Section 5

The notebook defines a manufactured streamfunction on the unit square.

psi = sin(pi x) sin(pi y) / pi

Velocity follows u = dpsi/dy and v = -dpsi/dx. Therefore

u = sin(pi x) cos(pi y)

v = -cos(pi x) sin(pi y)

Their divergence du/dx + dv/dy is zero. Out-of-plane vorticity is

omega = dv/dx - du/dy = 2 pi sin(pi x) sin(pi y).

The notebook independently checks these identities with centred finite differences in the interior. The checks have tolerances because discretization and roundoff introduce errors.

This analytic field has impermeable square walls. It does not impose the no-slip moving-lid conditions of a lid-driven cavity. Describe it as a manufactured demonstration field.

tf.stack([u, v, omega], axis=-1) gives shape (65, 65, 3). The last axis identifies the channel. All channels must refer to the same coordinates. sample[None, ...] adds a batch axis and produces (1, 65, 65, 3).

Twenty such snapshots have shape (20, 65, 65, 3). The first index selects a whole sample. In later CFD learning, train/test splits should reflect the intended physical prediction. Holding out a new geometry requires holding out that case's field, rather than a few neighbouring pixels.

Try it

Change grid size and repeat the derivative checks. Record interior divergence and vorticity errors. Explain why a consistent vorticity definition matters even though the network accepts any correctly shaped array.

When normalizing channels, state each scale and obtain learned normalization statistics from training data only. Velocity and vorticity need not have comparable ranges or the same units.

5. A loss needs a definition and a sample set

Notebook Section 6

For N matched scalar predictions, mean squared error is

MSE = sum of squared prediction-label differences / N.

The labels [1, 2, 3] and predictions [0.8, 2.2, 2.9] give squared errors 0.04, 0.04 and 0.01. Their sum is 0.09 and their mean is 0.03. Use this hand calculation to check TensorFlow.

For a velocity label measured in metres per second, MSE has units of squared velocity. Root mean squared error has velocity units. A quantity called loss without its definition and scale is incomplete.

A data loss measures disagreement with labels. A partial differential equation residual measures disagreement with a governing equation. A squared equation residual can contribute to a training objective, but it has a different interpretation from data MSE.

A physics-informed objective can combine data, equation and boundary terms. Each term needs a declared normalization. A small velocity-data MSE does not establish mass conservation or accurate pressure.

The sine example reports test MSE and relative L2 error. Relative L2 divides the error-vector norm by the reference-vector norm and multiplies by 100 for a percentage. This global normalization avoids dividing separately by reference values near the sine zeros. A nearly zero global norm would still require a different declared scale.

Try it

Multiply both labels and predictions by 10. Predict the MSE before running. It increases by a factor of 100. Add the same constant to both arrays and repeat. The residuals and MSE are unchanged.

Compare models on the same coordinates with the same label definition, units and scaling. Reducing a number by changing normalization does not establish improved physical accuracy.

6. Automatic differentiation records a computation

Notebook Section 7

Reference: TensorFlow automatic differentiation.

The notebook creates x0 as a trainable variable with value 3. Inside tf.GradientTape it calculates y0 = x0 squared. tape.gradient(y0, x0) returns 6.

A tape records supported TensorFlow operations executed while it is active. Trainable variables are watched automatically. An ordinary constant needs tape.watch when differentiating with respect to it.

The next example squares an unwatched constant and obtains None. Repeating with tape.watch(coordinate) gives 6. None means that no recorded differentiable path exists for the requested derivative. A connected derivative with value zero is different.

Converting an intermediate value to NumPy interrupts the recorded TensorFlow path. Keep TensorFlow operations inside the derivative calculation. Convert the final result for plotting or an independent check.

For prediction = a x + b, the MSE gradients are

dMSE/da = 2 mean((prediction - label) x)

dMSE/db = 2 mean(prediction - label).

The notebook calculates these formulas independently with NumPy. Before training, a and b are zero while labels follow 2 x + 0.5. Both gradients are negative. Subtracting a positive learning rate times the gradients increases the parameters.

Try it

Predict the gradient signs if all predictions exceed their positive labels. Check one small update. Calculate the loss again after updating rather than interpreting the loss at the previous parameter values.

The tape differentiates the computation you wrote. Labels, units, boundary conditions and the selected physical model still need independent scrutiny.

7. Understand one update before running many

Notebook Section 8

Reference: Gradient descent (SGD).

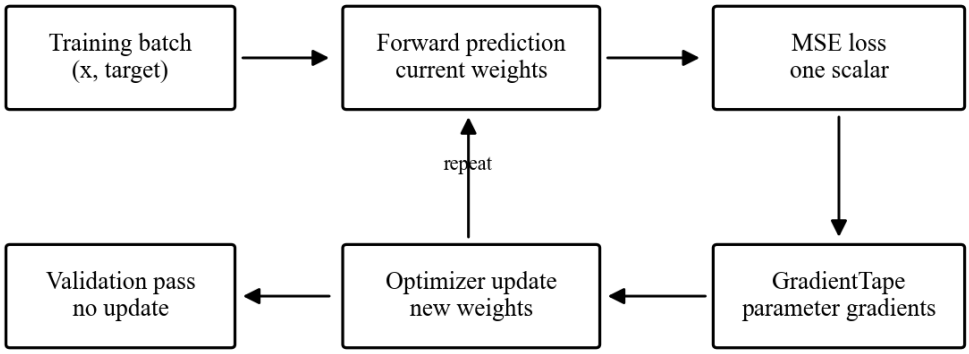
Gradient descent subtracts a scaled gradient.

a_new = a_old - learning_rate times dMSE/da

b_new = b_old - learning_rate times dMSE/db

The notebook starts a and b at zero and fits 50 points from the known relation 2 x + 0.5. Gradients are recalculated after every update. The history stores post-update MSE, so its last point corresponds to the printed final parameters.

The known coefficients provide a stronger check than a decreasing curve. Compare the predicted line with the reference and check both a and b.



At epoch end, evaluate train and validation with the same fixed weights.

One training update

The worked learning rate 0.5 suits this scaled quadratic problem. Changing the x range changes the objective's curvature and can change the stable step-size range.

The separate scalar diagnostic uses L(w)=(w-2) squared. Writing e=w-2 gives e_new=(1-2 eta) e_old, so the step-to-step loss ratio is (1-2 eta) squared whenever the old loss is nonzero. This exact result explains how an overly large step can increase loss. It does not give a universal stability limit for a network using Adam.

Try it

Change the target relation to 3 x - 0.2. Predict the coefficients, then train from a fresh initialization. Increase the learning rate in a separate experiment and identify when post-update loss first increases or becomes non-finite.

Keep update count and coordinates fixed when comparing rates. Changing target and rate together cannot isolate the effect of either.

8. A dense network builds a nonlinear function

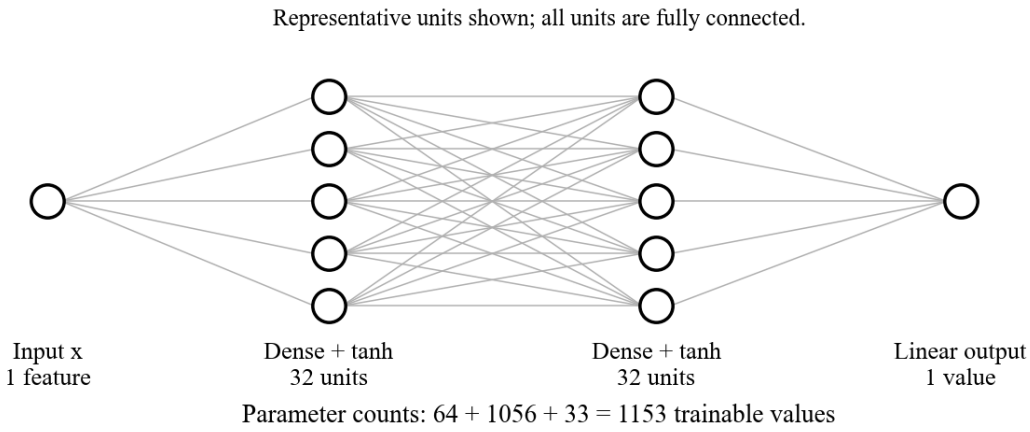
Notebook Section 9, model construction

Reference: The Keras Sequential model.

A dense layer forms weighted sums and adds a bias per output unit. A layer with d inputs and h outputs has d times h plus h parameters.

The hidden activation transforms these sums. The baseline uses two tanh hidden layers and a linear output. A linear output can represent both signs of sin(2 pi x).

The architecture 1 -> 32 -> 32 -> 1 has 64 parameters in its first layer, 1056 in its second and 33 in its output layer. The total is 1153. Compare this arithmetic with model.count_params().



Baseline dense network

Keras supplies Sequential and Dense. TensorFlow evaluates their operations and gradients. model.compile chooses optimizer and loss. model.fit performs training. model(x, training=False) evaluates the current weights without updating them.

Width counts the units in a hidden layer. Depth counts successive hidden layers. HIDDEN_UNITS=(32, 32) specifies two hidden layers of width 32. A tuple (16,) specifies one hidden layer.

Increasing width or depth changes capacity and optimization. A sequence of linear dense layers remains affine, so depth alone cannot accurately represent the full sine curve.

Try it

Calculate the count for 1 -> 16 -> 16 -> 1 before building it. The answer is 321. Compare with a single hidden layer of width 32. The parameter counts differ even when the widest layer has the same width.

A larger parameter count alone does not identify which model will achieve the lowest validation error within a fixed training budget.

9. Training, validation and test have different jobs

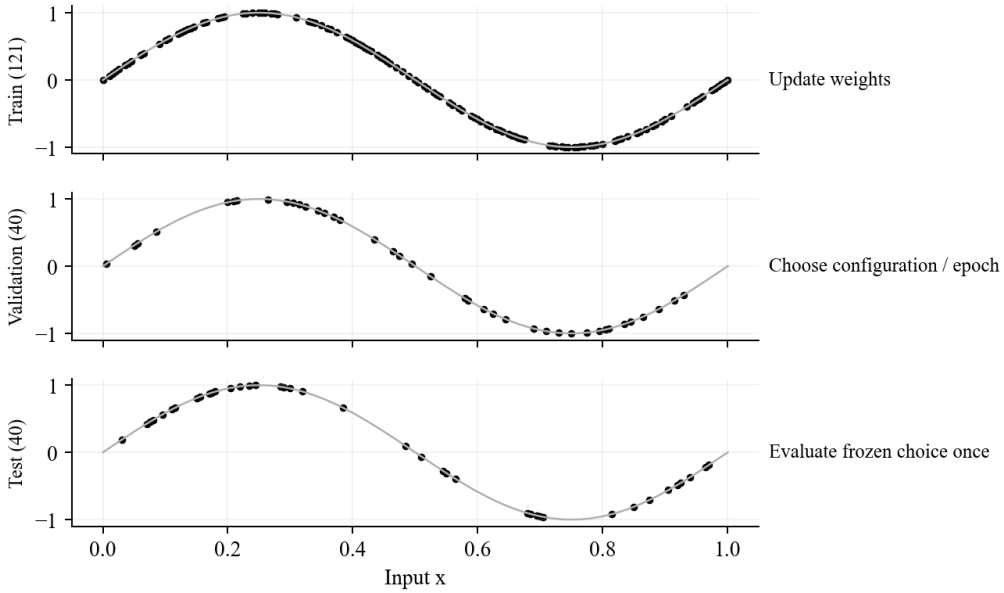
Notebook Section 9, data construction

Reference: Keras training and evaluation.

The baseline uses 121 training points, 40 validation points and 40 test points. Separate coordinate pools prevent overlap between roles.

Training points contribute gradients. Validation points assess the model during development and select stopping time or architecture. Test points assess the frozen choice. Selecting an architecture using test error makes those test points model-selection data.

The target is sin(2 pi x) on [0, 1]. The task measures interpolation at held-out coordinates from a known analytic function. It does not test a new Reynolds number, geometry or coordinate outside [0, 1].



Disjoint data roles

N_TRAIN, N_VAL and N_TEST request counts. make_sine_data uses disjoint, independently ordered pools with nested prefixes. Changing a training count preserves the validation and test coordinates. Training includes x=0 and x=1, enclosing the held-out points. N_TRAIN supports 2-2000 and N_VAL/N_TEST support 1-2000.

Nested training sets make a count study easier to interpret because the small set is contained in the large one. Inspect their spatial coverage as well as the count. Adding points changes where labels constrain the fit.

Forty points near x=0 do not cover a sine wave like forty points across the interval. Likewise, many nearby pixels in one field do not provide many independent flow cases.

Try it

Print all three shapes and check the overlap assertions. Repeat with N_TRAIN=30 while keeping validation and test counts fixed. Verify that only the training coordinates change.

For a later CFD generalization study, state which complete physical cases are held out. A random coordinate split and a held-out-geometry split answer different engineering questions.

10. Read the training configuration

Notebook Section 9, experiment controls

Reference: Keras Adam settings.

Edit the named settings, rerun their defining cell and create a fresh experiment. The helper resets the model and random seed.

Control	Baseline
N_TRAIN / N_VAL / N_TEST	121 / 40 / 40
HIDDEN_UNITS	(32, 32)
ACTIVATION	"tanh"
LEARNING_RATE	0.01
BATCH_SIZE	32
MAX_EPOCHS	300
SEED	42

An epoch is a pass through training data. A mini-batch is a subset used for an update. With N points and batch size B, a conventional pass uses ceil(N/B) batches, including a possible smaller final batch. Here, 121 samples and batches of 32 mean four updates per full epoch.

Increasing N_TRAIN with the same epoch cap therefore changes information and update count. Record both effects.

Adam uses estimates based on past gradients to adapt parameter updates. LEARNING_RATE still controls an overall step scale. The worked rate is specific to this small, scaled problem.

Early stopping monitors validation loss. restore_best_weights can return weights from an earlier epoch than the last point shown. Report the actual epoch count and selected validation result.

run_experiment returns a model, data, history, end_epoch_history and summary. The summary includes counts, parameter count, epochs_run, updates_run, train_mse, val_mse and fit_seconds. Its losses describe the restored model.

Try one controlled comparison

```
small = run_experiment(
    "train30", n_train=30, n_val=40,
    n_test=40, seed=42)
print(small["summary"])
```

Change one factor at a time. Keep coordinates and training controls fixed where possible. A fixed seed improves repeatability within an environment, but model rankings can change with initialization. Use validation-only repeats with declared seeds before drawing a broad conclusion.

11. Compare training and validation fairly

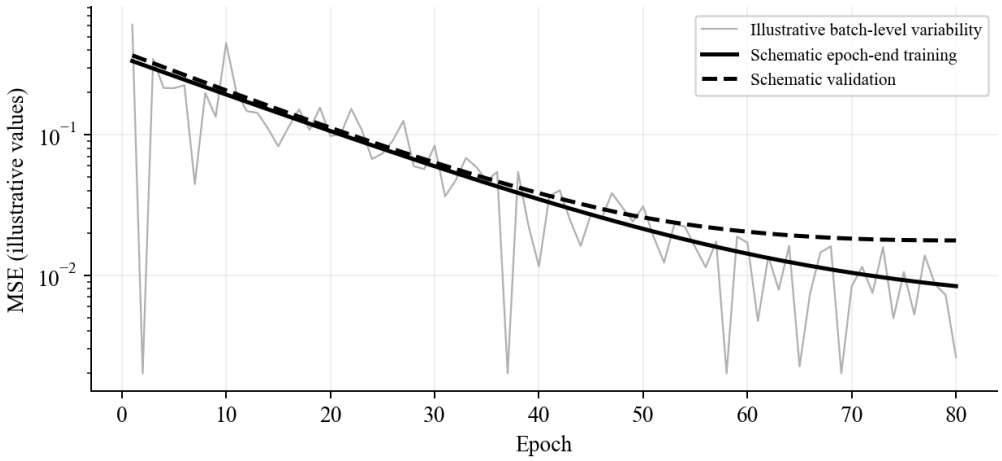
Notebook Section 9, loss histories

Reference: Keras explanation of training and validation loss.

Keras history["loss"] aggregates losses during an epoch while weights change between batches. Validation loss is normally calculated after the epoch using that epoch's weights.

The curves therefore differ in both data and evaluation timing. Validation can be lower than logged training loss with fully disjoint data.

The notebook also records end_epoch_history["train_mse"] and end_epoch_history["val_mse"]. Both use the same end-of-epoch weights with training=False. This removes the online-accumulation difference.



Schematic curves, not measured notebook results

The baseline has no dropout or batch-normalization layers. Other models can behave differently during training and inference. If regularization is added, loss can also include penalties beyond data MSE. Compare the same terms in the same mode.

Answer these questions

1. Do online and end-of-epoch training losses differ?
2. How does validation compare at the same weights?
3. Does the function profile reveal a related spatial error?

A finite validation set estimates performance at its particular coordinates. Its curve can fluctuate as weights change. One low point does not establish a broad improvement. Inspect neighbouring epochs, profiles and the stopping rule.

The last history entry and the restored model need not describe the same weights. To report the final selected model, use summary train_mse and val_mse, which are recomputed after restoration.

Read the actual arrays

```
same_weights = small["end_epoch_history"]
train_curve = same_weights["train_mse"]
val_curve = same_weights["val_mse"]
online_curve = small["history"].history["loss"]
```

Plot each against the matching epoch indices. Name the three curves so the reader can distinguish their definitions.

12. Investigate jumps in a loss curve

Notebook loss-diagnostic experiments

A jump is an observation. Investigate its cause through an intervention. First identify whether the horizontal axis counts batches, updates or epochs and what quantity was recorded.

Start from the same data and seed. Reduce only LEARNING_RATE. If large jumps shrink, that supports excessive step size as a contributor in this setup. It does not prove every jump had that cause.

Next increase BATCH_SIZE with other settings fixed. Mini-batch gradients use different subsets, which can pull updates in different directions. A larger batch often reduces this variability while changing updates per epoch.

Compare against updates or elapsed time as well as epochs when interpreting efficiency. A smooth full-batch curve can represent fewer updates.

Check data ordering and the helper's shuffle settings. Batches grouped by nearby x coordinates emphasise different parts of the sine wave. Record any ordering change.

Observation	**Controlled check**
Large spikes	Reduce learning rate
Batch fluctuations	Increase batch size
Epoch mismatch	Compare same-weight MSE
Non-finite values	Check scale and updates
Restart discontinuity	Fresh model and optimizer
Validation rises alone	Inspect point coverage

Use the scalar quadratic diagnostic to study a known loss ratio. Return to the network, where parameter interactions and Adam's state make the behaviour more complicated.

If the baseline has no jumps, report that. The optional learning-rate and batch-size experiment plans provide controlled conditions to investigate. Never add artificial spikes to an empirical curve.

Deliverable

Give the baseline, one lower-rate run and one changed-batch run. Show actual loss histories and profiles. Explain which intervention changed the behaviour and what remains uncertain.

13. Change training count and inspect the function

Student experiment A

Compare training counts 30, 121 and 484 with N_VAL=40 and N_TEST=40. Keep architecture, activation, seed, rate and stopping controls fixed.

Predict where sparse coverage could make the fit less accurate. After running, inspect coordinates, function profiles and same-weight training and validation histories.

Use validation results to compare runs. Keep test results reserved. Record the selected model's validation MSE, actual epochs, updates, parameter count and fit time.

Run	N_train	N_val
A1	30	40
A2	121	40
A3	484	40

```
counts = [30, 121, 484]
runs = [
    run_experiment(f"train{n}", n_train=n)
    for n in counts
]
for result in runs:
    print(result["summary"])
```

The optional experiment cell can run this plan by setting RUN_EXTRA_EXPERIMENTS=True and EXPERIMENT="training_count". Its separate EXPERIMENT_EPOCHS cap is 150 by default. State the cap used rather than comparing it silently with a 300-epoch baseline.

More points can improve spatial coverage and constrain the function. They also change updates per epoch. Discuss both effects. A lower training MSE with fewer samples can coexist with a larger validation error between sampled locations.

For a stronger data-efficiency study, compare a fixed update budget or fixed wall-clock budget. The worked lab uses an epoch cap and validation stopping, so describe its comparison accordingly.

Deliverable

Overlay the analytic function and three prediction profiles. Show training locations and validation histories. Explain one spatial improvement or deterioration supported by the figures. Retain the experiment summaries as a table.

14. Change validation and test counts thoughtfully

Student experiment B

Keep a trained model frozen and evaluate its validation MSE on 10, 40, 160 and 640 coordinates from the validation pool. The notebook's validation_size_audit performs this estimator check without retraining.

A larger validation count does not add gradient updates. It changes the locations used to estimate error. Small sets can miss a narrow region of poor prediction.

A second experiment can rerun early stopping using different N_VAL values. This can change which epoch appears best and therefore the restored model. Distinguish this selection effect from the frozen-model estimator effect.

Comparison	**Keep fixed**
Validation estimate	Model weights and train set
Early-stop selection	Train set and initialization
Test estimate	Final model and all choices

```
few_val = run_experiment(
   "val10", n_train=121, n_val=10)
more_val = run_experiment(
   "val40", n_train=121, n_val=40)
```

Compare the summary validation MSE and epochs_run, then inspect the profiles. The two reported MSE values use different evaluation sets, which must be acknowledged.

Choose N_TEST before the final test cell. Changing it after freezing the model changes the error estimator, not the learned network. Do not use that feedback to tune the model.

Deliverable

Explain whether the validation-count study changed primarily the estimate or the selected model. State how a small set could miss spatial error.

A count sweep does not automatically supply a confidence interval. Nested evaluation sets are dependent. Formal uncertainty assessment requires a declared resampling or independent-sampling design.

15. Make the network shallower, wider or deeper

Student experiment C

With fixed data, compare (32,), (32, 32) and (32, 32, 32, 32). Hold activation and training settings fixed.

These choices change depth and parameter count together. They test model configurations under a declared budget. They do not isolate depth independently of capacity.

For width, compare (8, 8), (32, 32) and (64, 64). For depth, hold width fixed while varying hidden-layer count. Record parameter counts in both studies.

```
shallow = run_experiment(
   "shallow", hidden_units=(32,))
deep = run_experiment(
   "deep", hidden_units=(32, 32, 32, 32))
```

A larger network represents a wider function family but may require more updates, different rates or more data. A worse result at the same epoch cap can reflect optimization difficulty.

Use fresh weights and optimizer state for each run. Continuing a previously trained model answers a different question because it benefits from earlier updates.

Evidence	**Question**
Profile	Where does the fit change?
Train MSE	Does it fit the labels?
Validation MSE	Does it fit between points?
Epochs and time	What did the result cost?
Parameter count	How much capacity changed?

Compare architectures using validation. Freeze the choice before the final test.

Deliverable

Provide function profiles and validation histories for the shallow, baseline and deep models. Describe the best and worst spatial regions. Do not replace physical profiles with bars of aggregate errors.

A shallow network matching the baseline supports a conclusion about this smooth one-dimensional task. It does not establish sufficiency for high-dimensional fluid fields or other data distributions.

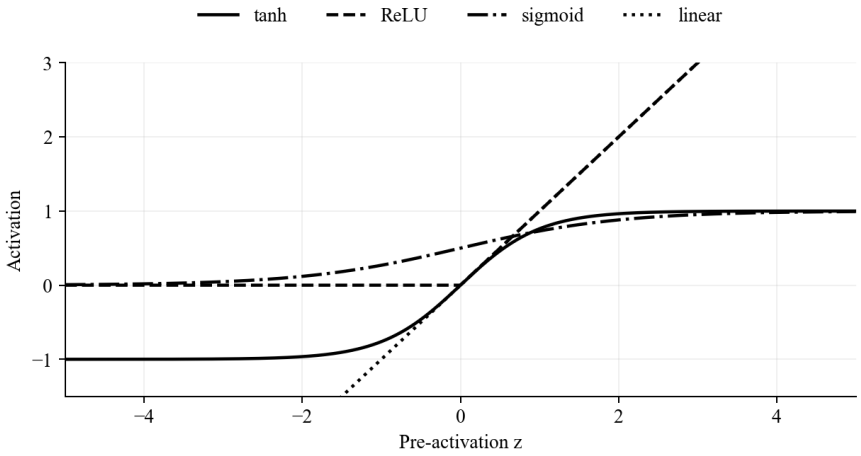
16. Change the activation and explain the result

Student experiment D

Reference: Keras activation functions.

ACTIVATION selects the hidden nonlinearity. Keep the output linear for the signed sine target.

tanh is smooth with output in [-1, 1]. Its slope decreases for large input magnitudes. Sigmoid is smooth with output in [0, 1] and can saturate. ReLU returns zero for negative input and the input itself for positive input. None specifies a linear layer.



Analytic activation profiles

Compare "tanh", "relu", "sigmoid" and "linear" with the same widths, seed and initial training controls.

```
relu_run = run_experiment(
    "relu", activation="relu")
linear_run = run_experiment(
    "linear", activation="linear")
```

Changing activation can change the usefulness of an initialization or learning rate. State that this first comparison fixes those choices.

A scalar-input ReLU network is piecewise affine. It can approximate sine values with many line segments. Its second derivative is zero within each segment and lacks a classical value at the corners. This matters for second-order physics residuals.

A small value error therefore does not make every activation equally suitable for a residual requiring curvature. Inspect first derivatives as well as function values.

Try it

Use "linear" in every hidden layer and derive why the resulting model is affine. Compare this prediction with its fitted profile. Then compare tanh and ReLU derivative profiles, especially near interval boundaries.

Deliverable

Submit function and first-derivative profiles, validation MSE and parameter counts. Explain one difference using activation shape and the learning evidence.

If a second derivative is disconnected or undefined in an implementation, report that explicitly. A substituted value is not an independently verified physical residual.

17. A derivative check differs from training a PINN

Notebook Section 10

The first nested-tape example uses the exact function sin(pi x). Its first derivative is pi cos(pi x), and its second derivative is -pi squared sin(pi x). Therefore the ordinary differential equation residual u" + pi squared sin(pi x) is zero analytically.

The notebook uses float64 and verifies derivatives and residuals against small tolerances.

The inner tape records the function and calculates the first derivative. The outer records that derivative calculation and obtains the second. Both explicitly watch coordinates.

The learned network approximates sin(2 pi x), whose derivative is 2 pi cos(2 pi x). Its residual uses the matching (2 pi) squared factor. The frequency must match the target.

The network was trained on function values. Its residual is a post-training diagnostic. A physics-informed neural network (PINN) would include equation and boundary information in its objective and require independent validation.

Pointwise derivatives

For a vector target, tape.gradient sums contributions. Each output row here depends only on its matching input row, so the result supplies the pointwise derivative. Layers coupling samples can require a Jacobian calculation.

Automatic differentiation differentiates the implemented composition. Finite differences approximate derivatives from nearby samples. Both need checks against known answers.

Try it

Change frequency consistently in the analytic function, first derivative and residual. Predict derivative amplitudes. Then change only the residual's frequency and observe the mismatch.

For the learned network, compare value and derivative errors near the interval ends. Accurate values alone do not guarantee accurate gradient or curvature.

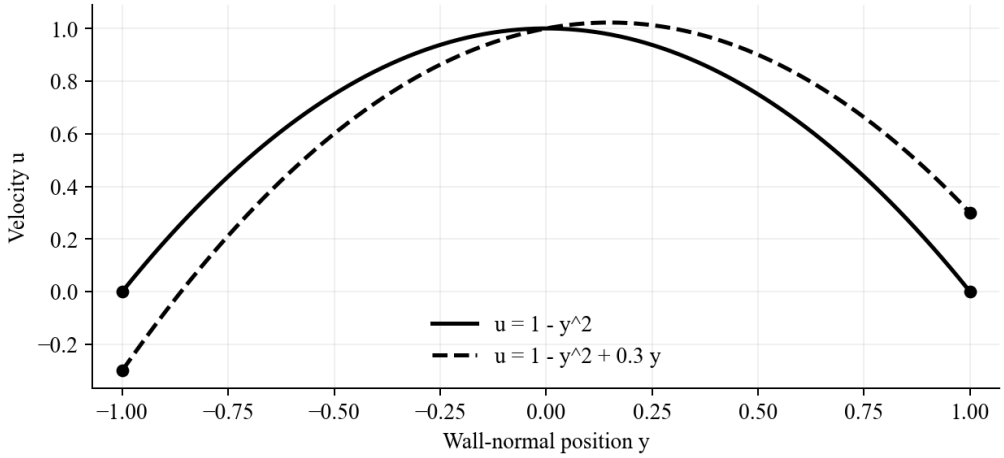
18. Equations and boundary conditions need separate checks

Notebook Section 10, channel example

The dimensionless steady, fully developed incompressible channel problem uses y in [-1, 1], u" + 2 = 0 and no-slip wall conditions u(-1)=u(1)=0.

The solution is u = 1 - y squared. The constant 2 represents the selected dimensionless pressure-gradient and viscosity ratio. This laminar example is separate from Week 2's rarefied-flow surrogate.

Adding 0.3 y leaves the second derivative unchanged. The modified profile satisfies the equation but has wall velocities -0.3 and +0.3.



Equation and boundary information

The notebook checks the equation residual and both wall values. For arbitrary c, u = 1 - y squared + c y gives u(-1)=-c and u(1)=c. The differential equation alone does not determine c.

Try it

Change c to -0.2 and 0.5. Predict wall values before running. Overlay the profiles against wall-normal coordinate and preserve the velocity signs.

Connection to physical learning

A loss can combine data, equation and boundary terms. A low total objective can hide a poorly satisfied component if scales or weights are unsuitable. Record separate terms and validate the predicted profile.

For a flow field, further checks can include conservation, pressure consistency and an independent numerical or measured reference. These analytic examples check mathematical identities and programming before the larger problems are attempted.

19. A CNN preserves spatial organisation

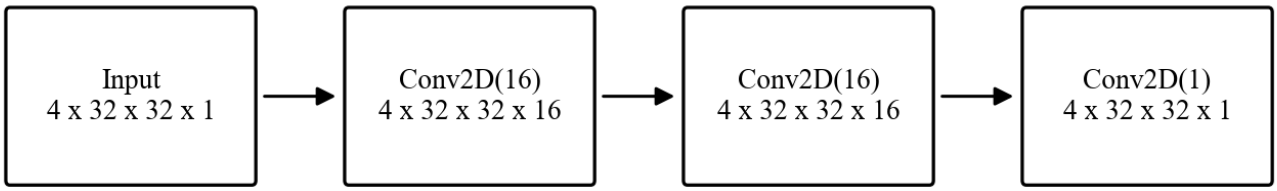
Notebook Section 11

Reference: TensorFlow Conv2D API.

A convolutional neural network (CNN) applies local learned filters at many grid locations with shared weights.

Input shape (4, 32, 32, 1) means four samples, a 32-by-32 grid and one channel. The first layer creates 16 channels, the second retains 16 and the final layer returns one.

With stride one and the specified "same" padding, the spatial dimensions remain 32 by 32. The output is (4, 32, 32, 1).



3 x 3 kernels, stride 1, same padding; 2625 trainable parameters.

CNN dimensions and channels

The inputs are random probes and the model is untrained. The output verifies a shape workflow and supplies no evidence of physical prediction accuracy.

A 3-by-3 convolution from one to 16 channels has 3 times 3 times 1 times 16 weights plus 16 biases, giving 160. This count does not grow with grid size because weights are shared.

The second layer has 3 times 3 times 16 times 16 plus 16 parameters, giving 2320. The output has 3 times 3 times 16 plus one, giving 145. The complete model has 2625 parameters.

Try it

Change the batch count from four to seven. Predict the output and parameter count. Then replace the input's one channel with three aligned channels. Update Input shape and recalculate the first layer's count.

Discuss how velocity and vorticity could form those channels when paired physical data are available.

A field image retains spatial layout. Units, coordinates and boundary information still supply its physical meaning. A trained field-to-field model also needs matched targets and an appropriate held-out evaluation.

20. Submission and interpretation checklist

Notebook Section 12

Your submission should let another student reproduce the experiment from a fresh kernel.

Include the manufactured scalar-field plot and explain its axes. Describe channel and batch dimensions. Give manual-fit coefficients and the known reference. Show the Keras profile, loss histories and final held-out evaluation.

Complete two controlled comparisons from the experiment menu, including one data-count and one network-design comparison. The remaining comparisons are optional extensions. Include the frozen-network validation-size audit and scalar loss-jump diagnostic. State unchanged controls and changes in updates or stopping time.

Questions to answer

1. Which samples updated weights?
2. Which selected stopping time and architecture?
3. When was the test evaluated?
4. Which training loss was compared with validation?
5. Which intervention explained a jump, if any?
6. Where did the profile improve or deteriorate?
7. Did value and derivative accuracy agree?
8. What did the channel and CNN checks verify?

Avoid attributing a difference to depth if parameter count and budget changed too. Explain which intervention supports the claim.

Symptom	First check
Missing TensorFlow	Active kernel and installation
Missing name	Run definitions from the top
Unexpected MSE	Label and prediction shapes
None gradient	Watched input and recorded path
Dtype error	Matching explicit dtypes
Non-finite loss	Scale and learning rate
Old Colab text	Reopen current source

API references

Read the official API explanations after the worked examples.

- TensorFlow tensors
- Automatic differentiation
- Keras training and evaluation
- Keras FAQ on training and validation loss